Supplementary Information: Reviving, reproducing, and revisiting Axelrod's second tournament

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Overview. In this paper, we focus on reproducing Axelrod's second tournament [1]. The **Supplementary Information** provides additional information and results that complement the main text. Namely, it is organized as follows.

- **Section S1** lists all 63 original submissions, including each strategy's author and the name of its Fortran function in the `Axelrod_fortran` library.
- **Section S2** reports how the original strategies rank across multiple tournaments considered in the paper, including our faithful reproduction, noisy variants, the Stewart and Plotkin tournament, and `Axelrod-Python` tournaments.
- **Section S3** provides the complete roster and ranks for the Stewart and Plotkin tournament [2], which now includes all original strategies.
- **Section S4** points to the full list of strategies and ranks for the comprehensive `Axelrod-Python` tournament (including the original submissions), available online.
- **Section S5** details the linear-model analysis using recursive feature elimination to identify subsets of strategies that best predict average scores. We show the resulting R^2 as a function of the number of strategies included (Figure 1).
- **Section S6** shows the original Fortran source code of `k42r` and summarizes its behavior.

S1 List of strategies from Axelrod's second tournament

Here we provide a complete list of the 63 strategies that participated in Axelrod's second tournament. The list includes the originally reported rank of each strategy, its author, and the name of its Fortran function implementation in the `Axelrod_fortran` library.

1. k31r - Original rank: 23. Authored by Gail Grisell
2. k32r - Original rank: 10. Authored by Charles Kluepfel
3. k33r - Original rank: 59. Authored by Harold Rabbie
4. k34r - Original rank: 52. Authored by James W Friedman
5. k35r - Original rank: 11. Authored by Abraham Getzler
6. k36r - Original rank: 63. Authored by Roger Hotz
7. k37r - Original rank: 37. Authored by George Lefevre
8. k38r - Original rank: 34. Authored by Nelson Weiderman
9. k39r - Original rank: 25. Authored by Tom Almy
10. k40r - Original rank: 35. Authored by Robert Adams
11. k41r - Original rank: 7. Authored by Herb Weiner
12. k42r - Original rank: 3. Authored by Otto Borufsen
13. k43r - Original rank: 39. Authored by R D Anderson
14. k44r - Original rank: 5. Authored by William Adams
15. k45r - Original rank: 50. Authored by Michael F McGurrin
16. k46r - Original rank: 14. Authored by Graham J Eatherley
17. k47r - Original rank: 16. Authored by Richard Hufford
18. k48r - Original rank: 53. Authored by George Hufford
19. k49r - Original rank: 4. Authored by Rob Cave
20. k50r - Original rank: 54. Authored by Rik Smoody
21. k51r - Original rank: 18. Authored by John William Colbert
22. k52r - Original rank: 48. Authored by David A Smith
23. k53r - Original rank: 45. Authored by Henry Nussbacher
24. k54r - Original rank: 58. Authored by William H Robertson
25. k55r - Original rank: 42. Authored by Steve Newman
26. k56r - Original rank: 38. Authored by Stanley F Quayle
27. k57r - Original rank: 31. Authored by Rudy Nydegger
28. k58r - Original rank: 21. Authored by Glen Rowsam
29. k59r - Original rank: 40. Authored by Leslie Downing
30. k60r - Original rank: 6. Authored by Jim Graaskamp and Ken Katzen
31. k61r - Original rank: 2. Authored by Danny C Champion
32. k62r - Original rank: 51. Authored by Howard R Hollander
33. k63r - Original rank: 57. Authored by George Duisman
34. k64r - Original rank: 17. Authored by Brian Yamachi
35. k65r - Original rank: 47. Authored by Mark F Batell
36. k66r - Original rank: 20. Authored by Ray Mikkelson
37. k67r - Original rank: 27. Authored by Craig Feathers
38. k68r - Original rank: 12. Authored by Fransois Leyvraz
39. k69r - Original rank: 29. Authored by Johann Joss
40. k70r - Original rank: 32. Authored by Robert Pebly
41. k71r - Original rank: 60. Authored by James E Hall
42. k72r - Original rank: 13. Authored by Edward C White Jr
43. k73r - Original rank: 41. Authored by George Zimmerman
44. k74r - Original rank: 61. Authored by Edward Friedland
45. k75r - Original rank: 8. Authored by Paul D Harrington
46. k76r - Original rank: 46. Authored by David Gladstein

47. k77r - Original rank: 55. Authored by Scott Feld
48. k78r - Original rank: 19. Authored by Fred Mauk
49. k79r - Original rank: 26. Authored by Dennis Ambuehl and Kevin Hickey
50. k80r - Original rank: 36. Authored by Robyn M Dawes and Mark Batell
51. k81r - Original rank: 43. Authored by Martyn Jones
52. k82r - Original rank: 49. Authored by Robert A Leyland
53. k83r - Original rank: 15. Authored by Paul E Black
54. k84r - Original rank: 9. Authored by T Nicolaus Tideman and Paula Chieruzz
55. k85r - Original rank: 33. Authored by Robert B Falk and James M Langsted
56. k86r - Original rank: 28. Authored by Bernard Grofman
57. k87r - Original rank: 44. Authored by E E H Schurmann
58. k88r - Original rank: 22. Authored by Scott Appold
59. k89r - Original rank: 56. Authored by Gene Snodgrass
60. k90r - Original rank: 24. Authored by John Maynard Smith
61. k91r - Original rank: 30. Authored by Jonathan Pinkley
62. k92r - Original rank: 1. Authored by Anatol Rapoport
63. krandomc - Original rank: 62. Authored by None

S2 Original submissions and their tournament rankings

For the original strategies, we evaluate performance in the following tournaments:

- **Reproduced tournament:** A faithful reproduction of Axelrod's second tournament.
- **Noisy tournament:** The original tournament with a noise (error) probability of 0.01.
- **Stewart and Plotkin tournament [2]:** A tournament with the original submission and the submissions used by Stewart & Plotkin.
- **`Axelrod-Python` tournament:** A tournament including all strategies from the `Axelrod-Python` library and the original submissions.
- **`Axelrod-Python` tournament with 1% noise:** As above, with 1% noise.
- **`Axelrod-Python` tournament with 5% noise:** As above, with 5% noise.

Table 1 shows the strategies and their ranks across these tournaments.

S3 Full list of strategies and ranks of Stewart and Plotkin tournament

The full list of strategies and their ranks in the Stewart and Plotkin tournament [2], which now includes the original strategies, is shown in Table 2.

S4 Full list of strategies and ranks for `Axelrod-Python` tournament

The full list of strategies and their ranks in the `Axelrod-Python` tournament, which now includes the original strategies, can be found online at.

	Original Rank	Reproduced Rank	Reproduced Rank (Noise 1%)	S & P tournament Rank	AXL tournament Rank	AXL (Noise 1%) tournament Rank	AXL (Noise 5%) tournament Rank
k92r	1	1	28	4	57	70	69
k61r	2	12	44	8	85	126	156
k42r	3	2	8	2	16	25	21
k49r	4	4	12	5	30	32	23
k44r	5	3	16	7	46	37	26
k60r	6	9	11	16	19	31	25
k41r	7	8	13	10	35	33	40
k75r	8	5	20	6	110	42	31
k84r	9	7	21	15	47	43	76
k32r	10	6	18	12	23	40	34
k35r	11	11	36	14	87	95	89
k68r	12	10	41	9	66	117	139
k72r	13	13	32	13	78	77	38
k46r	14	14	26	11	74	63	77
k83r	15	15	35	17	106	92	83
k47r	16	16	4	22	84	16	5
k64r	17	17	29	18	92	71	46
k51r	18	19	54	30	164	167	158
k78r	19	20	33	37	145	80	61
k66r	20	18	19	19	71	41	19
k58r	21	21	17	23	25	38	43
k88r	22	22	43	25	126	122	113
k31r	23	23	39	24	127	107	73
k90r	24	26	38	27	132	104	71
k39r	25	27	34	21	130	81	20
k79r	26	24	23	26	88	52	54
k67r	27	25	37	28	146	103	33
k86r	28	30	15	33	67	36	49
k69r	29	31	42	43	144	121	48
k91r	30	28	27	29	89	67	52
k57r	31	29	49	40	115	147	182
k70r	32	35	46	41	104	129	149
k85r	33	37	24	47	20	57	101
k38r	34	33	9	34	51	26	7
k40r	35	32	14	38	27	34	80
k80r	36	34	3	31	22	10	3
k37r	37	42	30	44	113	73	57
k56r	38	41	6	36	31	21	12
k43r	39	38	5	32	79	18	6
k59r	40	40	7	35	32	23	11
k73r	41	44	22	42	101	44	50
k55r	42	43	10	39	36	27	15
k81r	43	45	2	45	43	6	14
k87r	44	39	31	46	26	74	124
k53r	45	46	45	49	99	128	128
k76r	46	36	53	48	155	166	155
k65r	47	48	1	50	28	5	18
k52r	48	47	50	51	169	151	100
k82r	49	50	25	56	148	58	28
k45r	50	49	52	55	163	159	115
k62r	51	51	51	57	173	158	99
k34r	52	52	40	58	83	111	119
k48r	53	53	48	60	151	141	94
k50r	54	54	60	62	240	235	249
k77r	55	55	47	64	170	131	45
k89r	56	56	58	63	175	200	191
k63r	57	58	61	67	230	240	248
k54r	58	57	57	65	191	191	143
k33r	59	59	55	66	166	178	145
k71r	60	60	59	68	219	222	220
k74r	61	61	56	69	190	186	120
krandomc	62	62	63	71	260	252	255
k36r	63	63	62	72	247	251	243

Table 1: **Ranks of the original strategies across different tournaments.**

	Mean Score	Rank	Mean Cooperation Rate	Reproduced Mean Cooperation Rate	S. and P. Mean Cooperation Rate
ZD-GTFT-2: 0.25, 0.5	2.834	1	0.926	-	0.855
k42r	2.825	2	0.903	0.916	-
GTFT: 0.33	2.819	3	0.941	-	0.893
k92r	2.812	4	0.888	0.922	0.731
k49r	2.787	5	0.879	0.892	-
k75r	2.786	6	0.782	0.802	-
k44r	2.781	7	0.866	0.882	-
k61r	2.773	8	0.945	0.954	-
k68r	2.768	9	0.904	0.917	-
k41r	2.761	10	0.847	0.865	-
k46r	2.758	11	0.939	0.948	-
k32r	2.758	12	0.849	0.873	-
k72r	2.756	13	0.915	0.924	-
k35r	2.747	14	0.863	0.888	-
k84r	2.745	15	0.861	0.882	-
k60r	2.735	16	0.814	0.844	-
k83r	2.725	17	0.922	0.935	-
k64r	2.710	18	0.876	0.884	-
k66r	2.702	19	0.861	0.866	-
Hard Tit For 2 Tats	2.701	20	0.926	-	0.807
k39r	2.700	21	0.671	0.676	-
k47r	2.696	22	0.709	0.723	-
k58r	2.694	23	0.846	0.852	-
k31r	2.690	24	0.908	0.920	-
k88r	2.684	25	0.851	0.861	-
k79r	2.678	26	0.864	0.875	-
k90r	2.676	27	0.943	0.964	0.844
k67r	2.676	28	0.743	0.749	-
k91r	2.662	29	0.889	0.873	-
k51r	2.646	30	0.892	0.905	-
k80r	2.641	31	0.813	0.809	-
k43r	2.641	32	0.673	0.666	-
k86r	2.639	33	0.803	0.811	-
k38r	2.639	34	0.865	0.874	-
k59r	2.635	35	0.889	0.891	-
k56r	2.635	36	0.890	0.891	-
k78r	2.634	37	0.720	0.755	-
k40r	2.625	38	0.742	0.746	-
k55r	2.620	39	0.903	0.903	-
k57r	2.620	40	0.799	0.811	-
k70r	2.596	41	0.752	0.769	-
k73r	2.595	42	0.761	0.764	-
k69r	2.591	43	0.672	0.702	-
k37r	2.590	44	0.778	0.791	-
k81r	2.589	45	0.774	0.779	-
k87r	2.586	46	0.760	0.766	-
k85r	2.584	47	0.713	0.726	-
k76r	2.576	48	0.636	0.680	-
k53r	2.568	49	0.730	0.728	-
k65r	2.559	50	0.718	0.715	-
k52r	2.545	51	0.820	0.838	-
Win-Stay Lose-Shift	2.544	52	0.809	-	0.808
Hard Tit For Tat	2.539	53	0.710	-	0.663
Cooperator	2.522	54	1.000	-	1.000
k45r	2.521	55	0.726	0.754	-
k82r	2.491	56	0.529	0.544	-
k62r	2.490	57	0.732	0.754	-
k34r	2.478	58	0.647	0.652	0.655
Hard Go By Majority	2.375	59	0.658	-	0.459
k48r	2.300	60	0.504	0.499	-
First by Joss: 0.9	2.280	61	0.501	-	0.345
k50r	2.248	62	0.929	0.930	-
k89r	2.212	63	0.504	0.498	-
k77r	2.165	64	0.327	0.332	-
k54r	2.118	65	0.483	0.467	-
k33r	2.108	66	0.424	0.411	-
k63r	2.096	67	0.502	0.502	-
k71r	2.032	68	0.166	0.158	-
k74r	1.881	69	0.171	0.174	-
ZD-Extort-2: 0.1111111111111111, 0.5	1.838	70	0.254	-	0.232
krandomc	1.698	71	0.500	0.500	0.500
k36r	1.520	72	0.092	0.092	-
Defector	1.472	73	0.000	-	0.000

Table 2: Stewart and Plotkin tournament full list of rankings.

S5 Linear regression analysis

Using recursive feature elimination (RFE) [3], we select subsets of strategies to include in a simple linear model to predict average score. We begin with the full set of strategies. We estimate how useful each strategy is for predicting average scores, remove the least useful one, refit the model, and repeat. In this way, for any chosen number of strategies to include, we obtain the corresponding best subset.

We run this procedure for subset sizes from 1 to 25. For each size, we keep the subset that gives the highest R^2, as shown in Figure 1. The estimated R^2 rises from about 0.75 to about 0.95 when moving from 1 to 5 included strategies, after which improvements are smaller.

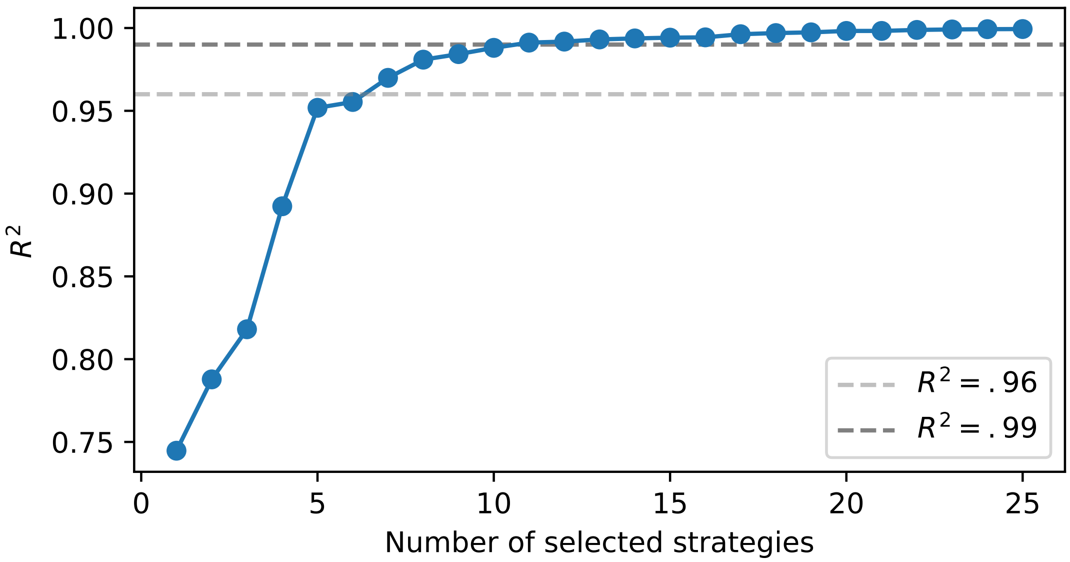

Figure 1: R^2 versus the *number of strategies to include* (subset size), obtained with recursive feature elimination using strategies. For each size, the shown value corresponds to the subset with the highest R^2 when predicting a strategy's average score (and thus its rank).

S6 Fortran source code for k42r

From inspection of the source code, the behaviour of the strategy can be described as follows (these descriptions are taken from the re-implementation in Python from [4], which has been verified to reproduce the original Fortran behaviour accurately):

This player keeps track of the opponent's responses to own behavior:

- `cd_count` counts: Opponent cooperates as response to player defecting.
- `cc_count` counts: Opponent cooperates as response to player cooperating.

The player has a defect mode and a normal mode. In defect mode, the player will always defect. In normal mode, the player obeys the following ranked rules:

1. If in the last three turns, both the player/opponent defected, then cooperate for a single turn.
2. If in the last three turns, the player/opponent acted differently from each other and they're alternating, then change next defect to cooperate. (Doesn't block third rule.)

3. Otherwise, do tit-for-tat.

Start in normal mode, but every 25 turns starting with the 27th turn, re-evaluate the mode. Enter defect mode if any of the following conditions hold:

- Detected random: Opponent cooperated 7-18 times since last mode evaluation (or start) AND less than 70% of opponent cooperation was in response to player's cooperation, i.e. `cc_count` / (`cc_count`+`cd_count`) < 0.7
- Detect defective: Opponent cooperated fewer than 3 times since last mode evaluation.

When switching to defect mode, defect immediately. The first two rules for normal mode require that the last three turns were in normal mode. When starting normal mode from defect mode, defect on first move.

References

[1] Axelrod, R. More Effective Choice in the Prisoner's Dilemma. *Journal of Conflict Resolution* **24**, 379–403 (1980).

[2] Stewart, a. J. & Plotkin, J. B. Extortion and cooperation in the Prisoner's Dilemma. *Proceedings of the National Academy of Sciences* **109**, 10134–10135 (2012).

[3] Guyon, I., Weston, J., Barnhill, S. & Vapnik, V. Gene selection for cancer classification using support vector machines. *Machine learning* **46**, 389–422 (2002).

[4] The Axelrod project developers. Axelrod-python/axelrod: v3.3.0. https://doi.org/10.5281/zenodo.836439 (2017).

```
       FUNCTION K42R(JPICK,MOVEN,ISCORE,JSCORE,RANDOM, JA)
C BY OTTO BORUFSEN
C TYPED FROM FORTRAN BY AX, 1/25/79
      DIMENSION MHIST(2,2)
      k42r=ja     ! Added 7/27/93 to report own old value
C INITIALIZE FIRST MOVE
      IF(MOVEN.NE.1)GOTO 20
         L3MOV=0
         L3ECH=0
         IDEF=0
         ICOOP=0
         IPICK=0
         DO 10 I=1,2
         DO 10 J=1,2
10       MHIST(I,J)=0
         GO TO 500
20    IF(MOVEN.EQ.2)GOTO 25
C UPDATE MOVE HISTORY
      MHIST(I2PCK+1,JPICK+1)=MHIST(I2PCK+1,JPICK+1)+1
25    IF(IDEF.EQ.0)GOTO 30
C OPPONENT HAS BEEN PROVED "RANDOM" OR
C "DEFECTIVE",I DEFECT FOR 25 MOVES
      K42R=1
      GO TO 100
30    IF(IPICK.EQ.0.OR.JPICK.EQ.0)GOTO 40
C MUTUAL DEFECTIONS ON LAST MOVE.
      L3MOV=L3MOV+1
      IF(L3MOV.LT.3)GOTO 50
C MUTUAL DEFECTION ON
C LAST THREE MOVES.
C I COOPERATE ONCE ON NEXT MOVE.
      K42R=0
      L3MOV=0
      L3ECH=0
      GOTO 100
C ONE (OR BOTH) COOPERATED ON LAST MOVE.
40    L3MOV=0
      IF(IPICK.EQ.JPICK)GOTO 45
      IF(JPICK.NE.I2PCK.OR.IPICK.NE.J2PCK)GOTO 45
C ECHO-EFFECT ON LAST MOVE.
      L3ECH=L3ECH+1
      IF(L3ECH.LT.3)GOTO 50
C ECHO-EFFECT ON LAST THREE MOVES.
C MY NEXT DEFECTION WILL BE SUBSTITUTED
C BY A COOPERATION.
      L3ECH=0
      L3MOV=0
      ICOOP=1
      GOTO 50
45    L3ECH=0
C PLAY 'TIT␣FOR␣TAT' AS MAIN RULE.
50    K42R=JPICK
100   IF(MOD(MOVEN-2,25).NE.0.OR.MOVEN.EQ.2)GOTO 650
C ON EVERY 25 MOVES:
C CHECK IF OPPONENT SEEMS TO BE
C 'RANDOM' OR 'DEFECTIVE'.
      IDEF=0
      JNCOP=MHIST(1,1)+MHIST(2,1)
C IS OPPONENT 'RANDOM'?
      IF(JNCOP.GT.17)GOTO 155
      IF(JNCOP.LT.8)GOTO 130
      IF(100*MHIST(1,1)/JNCOP.LT.70)IDEF=1
      GOTO 155
C IS OPPONENT 'DEFECTIVE'?
130   IF(JNCOP.LT.3)IDEF=1
155   DO 160 I=1,2
      DO 160 J=1,2
160   MHIST(I,J)=0
      IF(IDEF.EQ.0)GOTO 650
C OPPONENT SEEMS TO BE
C 'RANDOM' OR 'DEFECTIVE'.
C I DEFECT FOR NEXT 25 MOVES.
      ICOOP=0
      L3MOV=0
      L3ECH=0
      GOTO 600
C I COOPERATE.
500   K42R=0
      GOTO 650
C I DEFECT.
600   K42R=1
650   IF(ICOOP.EQ.0.OR.K42R.EQ.0)GOTO 660
         ICOOP=0
         K42R=0
660   I2PCK=IPICK
      J2PCK=JPICK
      IPICK=K42R
      RETURN
      END
```

Figure 2: Fortran Source code for original k42r strategy submitted to Axelrod's second tournament.